

Single-stranded DNA Binding Properties of the UvsX Recombinase of Bacteriophage T4: Binding Parameters and Effects of Nucleotides

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Bacteriophage T4 provides an important model for the biochemistry and genetics of DNA metabolism. Phage-encoded proteins conduct all essential steps of T4 DNA replication, repair, and recombination. Central to these three processes is the T4 UvsX protein, a member of the filamentous, ATP-dependent class of general recombination enzymes typified by the *Escherichia coli* RecA protein. Like RecA, UvsX forms presynaptic filaments on single-stranded (ss) DNA, which are the obligatory nucleoprotein intermediates in recombination. Aspects of the T4 presynaptic filament are explored by quantitative characterization of the UvsX-ssDNA interaction using an etheno-derivitized single-stranded DNA molecule, ϵ DNA, whose fluorescence is enhanced by UvsX binding. Studies with this model lattice show that UvsX exhibits a moderate level of cooperativity ($\omega = 100$) when binding to ϵ DNA with a binding-site size (n) equal to four nucleotide residues. Salt-stability studies of this complex reveal that the non-hydrolyzable ATP analog, ATP γ S, induces a high-affinity binding mode that is distinguishable from complexes formed with ADP or in the absence of a nucleotide cofactor. With this new information, both functional relationships between the UvsX and RecA recombinases, and implications for UvsX interactions with the other proteins of the T4 presynaptic filament (UvsY and gp32) may be further explored.

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Keywords: UvsX; recombination; ssDNA; bacteriophage T4; ATPase

Introduction

The UvsX gene product of bacteriophage T4 is a 40 kDa protein which catalyzes general genetic recombination. The enzymatic functions of the UvsX recombinase are similar to those of the well-characterized RecA protein of *Escherichia coli*, including single-stranded (ss) DNA-dependent ATPase and strand-transferase activities. Its mechanism includes the sequestration of ssDNA to form a nucleoprotein filament, synapsis of homologous double-stranded (ds) DNA, and subsequent 5' to 3' branch migration to form double-stranded heteroduplex DNA. (Yonesaki & Minagawa, 1985; Hinton & Nossal, 1986; Formosa & Alberts, 1986).

Abbreviations used: ssDNA, single-stranded DNA; ϵ DNA, etheno-modified single-stranded DNA; ATP γ S, adenosine 5'-O-(3-thiotriphosphate); KOAc, potassium acetate; PEP, phospho(enol)pyruvate; PK, pyruvate kinase; [salt], concentration of salt.

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Due to the tight coupling of recombination and replication in T4 infection, UvsX is essential for wild-type phage burst size as recombinational intermediates serve to initiate phage DNA replication during late stages of infection (reviewed by Kreuzer & Morrical, 1994). UvsX is also implicated in DNA-repair pathways dependent on recombination (reviewed by Kreuzer & Drake, 1994).

The key enabling step in catalytically active UvsX is the formation of the presynaptic filament, a helical nucleoprotein complex of UvsX wrapped around ssDNA. UvsX is capable of forming filaments on ssDNA in the absence of other proteins, however the T4 UvsY and gp32 proteins modulate this reaction strongly both *in vitro* and *in vivo* (Formosa & Alberts, 1986; Harris & Griffith, 1989; Kodadek *et al.*, 1989; Yonesaki & Minagawa, 1989; Morrical & Alberts, 1990). gp32, the single-stranded DNA binding protein of T4, interacts productively with UvsX at low concentrations to increase filament formation. Hindrance of UvsX loading at high gp32 (and salt) concentrations is overcome by the non-enzymatic accessory factor,

UvsY (Harris & Griffith, 1989; Kodadek *et al.*, 1989; Yonesaki & Minagawa, 1989; Morrical & Alberts, 1990). Protein-protein interactions among UvsX, UvsY and gp32 appear to moderate this process as well as the ssDNA interactions of each (Yonesaki & Minigawa 1989; Kodadek *et al.*, 1989; Jiang *et al.*, 1993, 1997; Yassa *et al.*, 1997). Possible rearrangement of the gp32-ssDNA complex by UvsY facilitates the assembly of the UvsX-ssDNA filament, displacing gp32 in the process (Jiang *et al.*, 1997; Beernink & Morrical, 1998; Sweezy & Morrical, 1998). Mutation in the UvsY protein is phenotypically indistinguishable from UvsX mutation *in vivo*, illustrating its essential role in T4 recombination (Cunningham & Berger, 1977; Melamede & Wallace, 1977).

While the enzymatic activities of UvsX protein resemble those of RecA protein at least qualitatively, quantitative aspects of UvsX-ssDNA interactions have yet to be determined. Such information is desirable since it offers a unique opportunity to compare detailed biochemical properties with the only other well-characterized general recombinase, RecA protein. Likewise, since UvsX interacts specifically with two other T4 ssDNA-binding proteins (UvsY and gp32) which modulate its activities, quantitative characterization of UvsX protein also offers the opportunity to examine the relationships between the ssDNA binding properties of a recombinase and its major accessory proteins. This is especially important since similar accessory activities have now been identified in both bacterial and yeast recombination systems, e.g., the RecO and RecR proteins of *E. coli* which enhance RecA-ssDNA filament formation (Umeza *et al.*, 1993; Umeza & Kolodner, 1994; Shan *et al.*, 1997), and the Rad52 and Rad55/57 proteins of yeast which appear to enhance Rad51-ssDNA filament properties (Sung, 1997a,b; Benson *et al.*, 1998; Shinohara & Ogawa, 1998; New *et al.*, 1998). Accordingly, here we present the first detailed biochemical analysis of the ssDNA-binding properties of the T4 UvsX protein, using an etheno-modified ssDNA (ϵ DNA) fluorescence enhancement assay as a quantitative model for UvsX-ssDNA interactions. Our findings include the following: UvsX protein binds to an infinite etheno-DNA (ϵ DNA) lattice with a binding-site size of four nucleotide residues per protein monomer ($n = 4$), with moderate cooperativity ($\omega = 100$), and with an intrinsic affinity (K) that is modulated by both cations and anions. In addition, salt-stability studies demonstrate the existence of a high-affinity ssDNA-binding mode of UvsX induced by the non-hydrolyzable ATP analog ATP γ S, whereas the product of UvsX-catalyzed ssDNA-dependent ATP hydrolysis, ADP, has no observable effect on UvsX-ssDNA binding. We discuss the implications of our results for the function of T4 presynaptic filaments, as well as for the functional similarities and differences within the RecA-like family of general recombinases.

Results

UvsX binding to ϵ DNA; binding-site size determination

The determination of the binding-site size parameter (n) of the UvsX protein on a model ϵ DNA lattice derived from M13mp19 was performed by the fluorescence-enhancement assay as described in Materials and Methods. Under salt and buffer conditions that permit tight, stoichiometric binding (namely 50 mM NaCl or KOAc in chloride or acetate-based buffer, respectively, see Figure 2), the UvsX protein was titrated onto fixed amounts of the ϵ DNA lattice in the absence of other nucleotide ligands (Figure 1). These forward titrations resulted in a relative fluorescence enhancement of approximately 2 to 2.5-fold, depending on the ϵ DNA batch used, when comparing fluorescence of the saturated *versus* unsaturated UvsX- ϵ DNA complex. Use of two different preparations of ϵ DNA resulted in slightly different degrees of fluorescence enhancement between the free ϵ DNA and UvsX-saturated complexes (data not shown), presumably due to varying levels of etheno derivatization of the ssDNA lattice; however, the break points indicating saturation of binding-sites were unaltered between ϵ DNA batches. The ratio of ϵ DNA concentration (in nucleotides) to the molar concentration of UvsX protein at the break point of each titration was used to determine the apparent binding-site size (n). At the three lattice concentrations examined (1.05 μ M, 2.1 μ M, and 3.2 μ M), the saturation points of UvsX protein were directly proportional to lattice concentration, indicating that over a threefold DNA concentration range, the binding-site size remained constant at approximately $n = 4$ nucleotide residues per UvsX monomer (Figure 1; Table 1). This result is in excellent agreement with previous site size estimates for UvsX-ssDNA complexes derived from electron microscopy and from ATPase assays (Griffith & Formosa, 1985; Kodadek *et al.*, 1989), and is also similar to n -values obtained for the *E. coli* RecA protein (Morrical & Cox, 1985; Menetski & Kowalczykowski, 1985).

Table 1 summarizes the n -values obtained from a number of such titrations, exemplified by Figure 1. Also shown in Table 1 are results of similar titrations (not shown) performed in the presence of the nucleotide ligands ADP, ATP γ S, and ATP (plus an ATP-regeneration system). As seen in Table 1, none of these nucleotide ligands of UvsX protein had a significant effect on the value of n ; under all nucleotide conditions and in both buffer systems (chloride and acetate), the apparent binding-site size remains 4.0($\pm$ 0.3) nucleotide residues per monomer. Therefore the binding-site size of UvsX protein on a single-stranded lattice is assumed to be invariant, and a parameter value of $n = 4$ is used in all subsequent calculations of UvsX- ϵ DNA affinity, cooperativity, and ionic parameters.

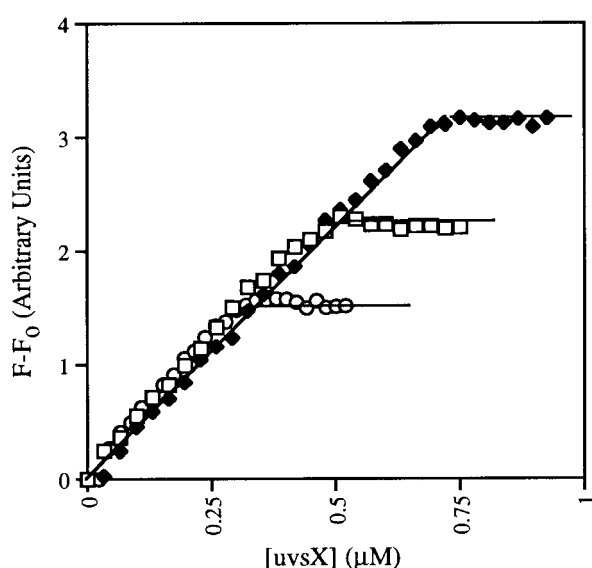


Figure 1. Forward titrations of ϵ DNA with UvsX protein under stoichiometric (tight) binding conditions. Titrations monitored by ϵ DNA fluorescence enhancement were carried out as described in Materials and Methods, in chloride buffer containing 50 mM NaCl. ϵ DNA concentrations were as follows: $\circ$, 1.05 μ M; $\square$, 2.10 μ M; $\blacklozenge$, 3.15 μ M. All data were corrected for intrinsic protein fluorescence, inner filter effects, and dilution as described in Materials and Methods. F_0 denotes the starting fluorescence of ϵ DNA before any protein addition, which was subtracted from the data in each titration. Units of the ordinate axis are arbitrary machine units of ratiometric sample-standard fluorescence. Titration endpoints, used to calculate the value of the binding-site size parameter n , were estimated from the intersections of lines drawn through the linear ascending and plateau portions of the titration courses as shown in the Figure.

The effects of nucleotides on the maximal ϵ DNA fluorescence enhancement ($F_{\max}$ value) for saturated UvsX- ϵ DNA were also tested; saturated complexes were examined in the presence of no nucleotide, ADP (200 μ M), ATP γ S (50 μ M), or ATP (100 μ M $\pm$ regenerating system). The results are summarized in Table 1. All three nucleotides produced negligible changes in $F_{\max}$ relative to the experiment with no nucleotide cofactor, after correction for inner filter effects of the nucleotides. This contrasts with observations made for the *E. coli* RecA protein (Menetski & Kowalczykowski, 1985), which indicated increases in $F_{\max}$ of RecA- ϵ DNA in the presence of ATP or ATP γ S, and a fluorescence decrease with the inclusion of ADP when compared to reactions lacking cofactor.

Preference of UvsX for etheno-modified *versus* unmodified ssDNA

Competition titrations were performed using equimolar concentrations of ϵ DNA and unmodified M13mp19 ssDNA (data not shown). The forward titrations, monitored by ϵ DNA fluorescence

enhancement, resulted in essentially identical titration curves with the break point occurring at the same UvsX concentration as seen in titrations of ϵ DNA alone. This result, obtained in both chloride and acetate buffer systems under stoichiometric binding conditions for UvsX- ϵ DNA, indicates the preference of UvsX for the etheno-modified over the unmodified ssDNA lattice. Thus far, a majority of other ssDNA-binding proteins which have been tested exhibit a similar preference for ϵ DNA, including *E. coli* RecA and the T4 UvsY and gp32 proteins (Sweezy & Morrical, 1997; Kowalczykowski *et al.*, 1981; Menetski & Kowalczykowski, 1985). Exceptions include the T4 gp59 and yeast RP-A proteins (Lefebvre & Morrical, 1997; Alani *et al.*, 1992).

Salt stability of the UvsX- ϵ DNA complex

The stabilities of pre-formed UvsX- ϵ DNA complexes were tested by monitoring fluorescence change during salt addition. These salt-back titrations examine three aspects of the UvsX- ϵ DNA interaction: (1) the identification of the salt concentrations where non-stoichiometric (weak) UvsX- ϵ DNA binding occurs, (2) contributing effects of anionic interactions on UvsX binding with chloride *versus* acetate counterions, and (3) the determination of nucleotide effects on the complex stability.

In all of these titrations, UvsX was incubated with a stoichiometric amount of ϵ DNA, assuming a binding-site size of four nucleotides per UvsX monomer in either the chloride or acetate buffer. Complex stability was tested by stepwise additions of concentrated NaCl or KOAc. The data are shown in Figure 2, and summarized in Table 2.

Table 1. Binding-site size and ϵ DNA fluorescence enhancement values for UvsX- ϵ DNA complexes in the presence and absence of nucleotide cofactors

Nucleotide cofactor	n (nucleotides) ^a	Fluorescence enhancement ($F_{\max}$) ^b
None	4.1 $\pm$ 0.2	2.1
ADP	4.0 $\pm$ 0.1	2.1
ATP γ S	4.1 $\pm$ 0.1	1.9
ATP ^c	4.0 $\pm$ 0.3	2.0
Composite Average:	4.0 $\pm$ 0.3	

Values of n (UvsX binding-site size on ϵ DNA) represent the averages ($\pm$ standard deviation) determined from three independent titration experiments such as those shown in Figure 1, involving two independent batches of ϵ DNA. The composite average shown is based on all 12 of these titrations. Exact reaction conditions were as described in Materials and Methods and in the legend to Figure 1.

^a Binding site size of UvsX protein on ϵ DNA in nucleotide residues per monomer.

^b Relative enhancement of ϵ DNA fluorescence by UvsX protein at saturation. Values shown here compare the same batch of ϵ DNA. A second, independent batch of ϵ DNA gave slightly higher levels of fluorescence enhancement at saturation, but the relative effects of nucleotide cofactors were the same with both ϵ DNA batches.

^c Mixtures containing ATP also contained a PEP/PK ATP-regenerating system as described in Materials and Methods.

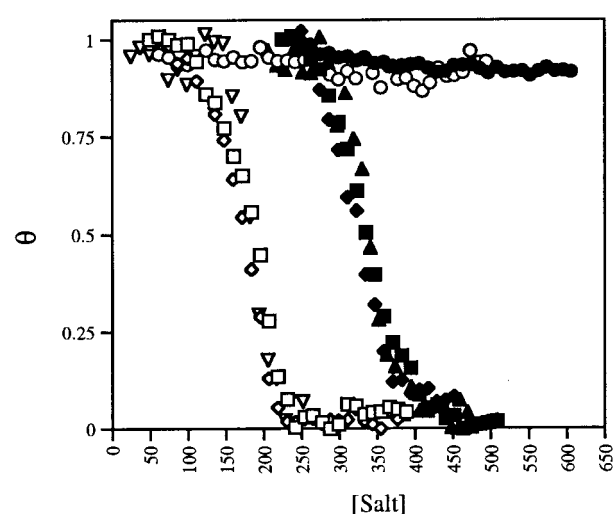


Figure 2. Salt-back titrations of pre-formed UvsX- ϵ DNA complexes in the presence and absence of nucleotide cofactors, monitoring complex dissociation by decreasing ϵ DNA fluorescence. The 1:1 stoichiometric complexes of UvsX protein and ϵ DNA were preincubated in chloride or acetate buffer, then titrated with increasing amounts of NaCl (open symbols) or KOAc (filled symbols), respectively, as described in Materials and Methods. The data were corrected for dilution, protein fluorescence, and the fluorescence of free ϵ DNA, then converted to fractional saturation of the lattice (θ) by normalizing the starting signal of each titration curve to a value of $\theta = 1$. Titration mixtures contained nucleotide cofactors as follows: $\square$ and $\blacksquare$, no cofactor; $\circ$ and $\bullet$, 100 μ M ATP γ S; $\diamond$ and $\blacklozenge$, 200 μ M ADP; ∇ and $\blacktriangle$, 100 μ M ATP plus the ATP-regenerating system as described in Materials and Methods.

The decrease in fluorescence observed with increased salt concentration is attributable to the dissociation of the UvsX- ϵ DNA complex. The salt range wherein UvsX partially dissociates from the ϵ DNA provides the range of non-stoichiometric binding. In the absence of nucleotide cofactors, this non-stoichiometric range falls between 100 mM and 230 mM NaCl, with a midpoint of approximately 180 mM NaCl. Identical results were obtained with KCl as a titrant (data not shown), indicating the absence of a specific monovalent cation effect on UvsX- ϵ DNA binding. The non-stoichiometric range for KOAc is 250 to 425 mM, with a midpoint of approximately 340 mM KOAc. Note that the greater stability of UvsX- ϵ DNA in the presence of acetate *versus* chloride counterions demonstrates an anion effect on this interaction, which is discussed in a later section.

The effects of nucleotide cofactors on the salt stability of UvsX- ϵ DNA are also shown in Figure 2 and summarized in Table 2, and reveal three striking features. First, the presence of the non-hydrolyzable ATP analog, ATP γ S, has a dramatic stabilizing effect on UvsX- ϵ DNA complexes. At an ATP γ S concentration of 100 μ M, no dissociation of UvsX- ϵ DNA was detected at salt concentrations as

high as 1.0 M NaCl and 1.3 M KOAc, respectively, the highest concentrations tested. Identical results were obtained at 50 μ M ATP γ S (data not shown). Since ATP γ S is not hydrolyzed once bound to the UvsX, it is surmised that this nucleotide analog traps UvsX in a high-affinity ssDNA-binding state, mimicking a condition that occurs transiently in the presence of a hydrolyzable nucleotide. This observation is analogous to the RecA model (Menetski & Kowalczykowski, 1985; Kowalczykowski *et al.*, 1994).

A second key feature is that ADP, the major nucleotide product of UvsX-catalyzed ssDNA-dependent ATP hydrolysis, has no observable effect on the salt stability of UvsX- ϵ DNA complexes. Salt-back titrations performed in the presence of either 200 μ M ADP (Figure 2) or of 500 μ M ADP (data not shown) each were essentially indistinguishable from titrations performed in the absence of nucleotide cofactors (Figure 2). In addition, an identical stabilizing effect of the acetate *versus* the chloride counterion was observed in the presence and absence of ADP (Figure 2). These results demonstrate that ADP has neither a stabilizing nor a destabilizing effect on UvsX- ϵ DNA interactions at equilibrium. This stands in sharp contrast to the situation for RecA- ϵ DNA complexes, where ADP has a marked, concentration-dependent destabilizing effect on RecA- ϵ DNA interactions (Menetski & Kowalczykowski, 1985). The absence of any ADP-mediated destabilization of the UvsX- ϵ DNA complex suggests differing modes of control for the formation and disruption of the UvsX-ssDNA filament than for its *E. coli* counterpart (see Discussion).

A third key feature involves the effects of ATP. Preliminary work (unpublished results) has estab-

Table 2. Midpoints of salt-back titrations monitoring the dissociation of pre-formed UvsX- ϵ DNA complexes in the presence and absence of nucleotide cofactors

Nucleotide cofactor	Salt	Titration midpoint (mM) ^a
None	NaCl	180
ADP	NaCl	180
ATP γ S	NaCl	>1000 ^b
ATP ^c	NaCl	180
None	KOAc	340
ADP	KOAc	340
ATP γ S	KOAc	>1300 ^d
ATP ^c	KOAc	340

Data derived from salt-back titrations such as those shown in Figure 2. Exact reaction conditions were as described in Materials and Methods and in the legend to Figure 2.

^a Estimated point at which a given salt-back titration curve crosses the half-saturation ($\theta = 0.5$) point.

^b No dissociation of UvsX- ϵ DNA was observed in the presence of ATP γ S at NaCl concentrations as high as 1.0 M, the highest concentration tested.

^c Mixtures containing ATP also contained a PEP/PK ATP-regenerating system as described in Materials and Methods.

^d No dissociation of UvsX- ϵ DNA was observed in the presence of ATP γ S at KOAc concentrations as high as 1.3 M, the highest concentration tested.

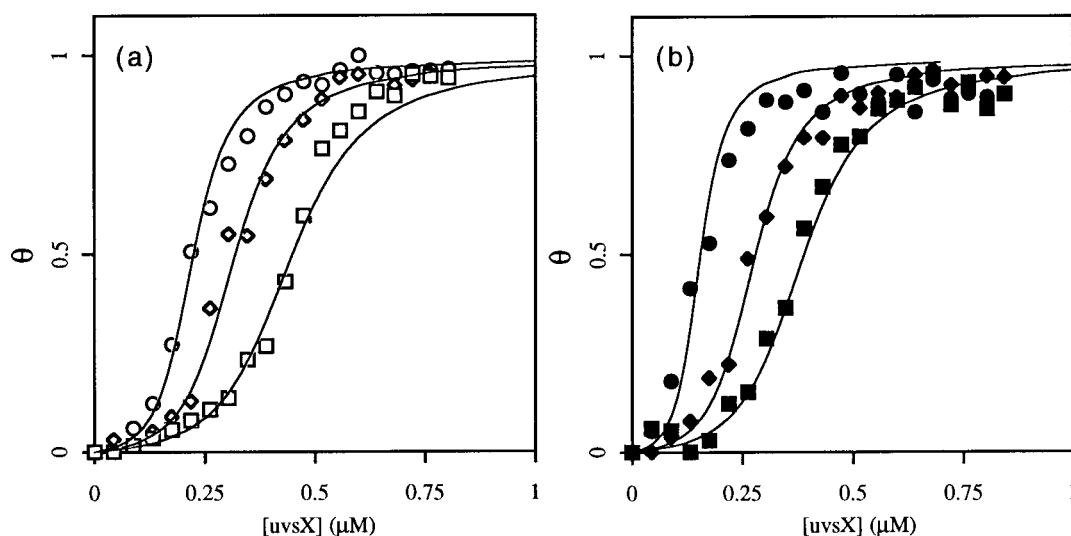


Figure 3. Forward titrations of ϵ DNA with UvsX protein under non-stoichiometric (weak) binding conditions. Fluorescence data were corrected for intrinsic protein fluorescence, inner filter effects, dilution, and the fluorescence of free ϵ DNA, then converted to fractional saturation of the lattice (θ) by normalizing to the plateau fluorescence value at saturation ($\theta = 1$) as described in Materials and Methods and the text. (a) Titrations performed in chloride buffer (without nucleotide cofactors) plus: $\circ$, 100 mM, $\diamond$, 125 mM NaCl; and $\square$, 150 mM NaCl, respectively. (b) Titrations performed in acetate buffer (without nucleotide cofactors) plus: $\bullet$, 250 mM KOAc; $\blacklozenge$, 300 mM KOAc; and $\blacksquare$, 350 mM KOAc, respectively. In both (a) and (b), continuous lines represent the visually “best fit” theoretical isotherms for each data set, generated from the equations of McGhee & von Hippel (1974, 1976) as described in the text. The K and ω parameters derived from these fits are listed in Table 3.

lished that an ϵ DNA lattice stimulates the ssDNA-dependent ATPase activity of UvsX protein to a similar extent as native ssDNA. Nevertheless, salt-back titrations of UvsX- ϵ DNA performed in the presence of ATP plus an ATP-regenerating system yielded results that were essentially identical with experiments performed with ADP or with no cofactor (Figure 2). Separate controls (not shown) performed with ATP plus or minus the regenerating system demonstrated that UvsX hydrolyzes ATP to ADP + Pi rapidly in these reaction mixtures, but that the ATP-regeneration system used (see Materials and Methods) maintains a very high ATP/ADP ratio throughout the course of these titrations. Thus, unlike ATP γ S, ATP appears to offer little stability advantage to UvsX- ϵ DNA, perhaps because ATP is turned over very rapidly by UvsX (k_{cat} 20 to 25-fold greater than that of the *E. coli* RecA protein; Kodadek, 1990). Thus an ATP-induced high-affinity ssDNA-binding mode of UvsX may be extremely transient, arguing for other factors necessary to stabilize the presynaptic filament (see Discussion). Note that these results contrast with those observed with the RecA protein, which in the presence of ATP exhibits a salt

stability intermediate between that seen with ATP γ S and with no cofactor (Menetski & Kowalczykowski, 1985).

Determination of binding parameters K and ω

The equilibrium affinity (K) and cooperativity (ω) parameters for UvsX- ϵ DNA interactions were determined by performing forward titrations of lattice with protein as described in Materials and Methods, using salt conditions which allow non-stoichiometric binding. These conditions were selected based on results obtained in the salt-back titrations (Figure 2 and Table 2). Non-stoichiometric binding was effectively limited to solutions containing moderate salt concentrations and either no nucleotide or ADP, since we were unable to observe dissociation of complexes in the presence of ATP γ S † . In addition, salt conditions were chosen that gave reasonable “plateau” fluorescence values at saturation to aid in the calculation of fractional lattice saturation; for NaCl this involved concentrations of 100 to 150 mM, and for KOAc this involved concentrations of 250 to 350 mM. Corrections of the data for dilution, inner filter effects, and protein fluorescence, plus the calculation of experimental fractional lattice saturation (θ) values were as described in Materials and Methods.

Figure 3 shows UvsX- ϵ DNA binding isotherms obtained in the absence of nucleotide cofactors at different NaCl (Figure 3(a)) and KOAc (Figure 3(b)) concentrations. Virtually identical results were obtained in the presence of 200 μM ADP (data not shown). All curves exhibited obvious sigmoidicity,

† Reactions employing ATP (plus the regenerating system) were judged inappropriate for these equilibrium experiments, since ssDNA-dependent ATP hydrolysis by UvsX protein sets up an obvious non-equilibrium situation, and since the possibility exists that the ATP substrate and ADP product may affect UvsX-ssDNA interactions differently as the binding density changes during the titration.

Table 3. Values of $K\omega$, K , and ω for UvsX- ϵ DNA interactions as functions of salt concentration

	[Salt] (mM)	$K\omega$ (M^{-1}) ^a	K (M^{-1}) ^b	ω ^c	$K_{Mg-corr}$ (M^{-1}) ^d
NaCl	100	9.5×10^6	9.5×10^4	100	2.7×10^5
	125	6.5×10^6	6.5×10^4	100	1.4×10^5
	150	4.3×10^6	4.3×10^4	100	7.5×10^4
KOAc	250	1.3×10^7	1.3×10^5	100	1.7×10^5
	300	7.2×10^6	7.2×10^4	100	8.9×10^4
	350	5.3×10^6	5.3×10^4	100	6.2×10^4

Data derived from forward, non-stoichiometric titrations such as those shown in Figures 3 and 4, performed in the absence of nucleotide cofactors. Essentially identical results were obtained in the presence of 200 μ M ADP (data not shown). Other reaction conditions were as described in Materials and Methods and in the legends to Figures 3 and 4.

^a Values derived from the midpoint of the titration (point at which titration reaches half-saturation, i.e., $\theta = 0.5$) as described in Materials and Methods.

^b Association constants derived from visual “best fits” of data to theoretical isotherms as described in the text and Materials and Methods.

^c Dimensionless cooperativity parameter derived from visual “best fits” of data to theoretical isotherms as described in the text and Materials and Methods.

^d Association constants corrected for the effects of Mg^{2+} as described in the text. These values are plotted in Figure 5.

indicative of positive cooperativity in the binding of UvsX to the ϵ DNA lattice under these weak, non-stoichiometric binding conditions (compare with the forward stoichiometric titrations in Figure 1, which display linear increases in fluorescence prior to saturation). The curves were each fitted visually to the equations of McGhee & von Hippel (1974, 1976), as described in Materials and Methods, to estimate the values of K and ω . The theoretical isotherms generated are shown as continuous lines in Figure 3, and the K and ω parameter values are listed for each salt concentration in Table 3.

A specific example of curve fitting and K and ω estimation is shown in Figure 4, for data obtained at 150 mM NaCl in the chloride buffer system. Here the product $K\omega$ is equal to $4.3 \times 10^6 M^{-1}$, a value derived from the midpoint of the titration as described (see Materials and Methods). Different theoretical isotherms (shown as broken lines) were generated by varying the values of K and ω while holding the product $K\omega$ constant, and the “best fit” theoretical isotherm (shown as the continuous line) was judged by visually fitting to the overlaid experimental data (Figure 4). Using this method, a reasonable fit of the data to an isotherm represented by parameters $K = 4.3 \times 10^4 M^{-1}$ and $\omega = 100$ was obtained. (All calculations assumed a binding-site size parameter of $n = 4$.) As shown in Figure 4, assuming a cooperativity parameter of $\omega = 100$ clearly gives a better fit than either a higher cooperativity case (e.g., $\omega = 250$) or a lower cooperativity case (e.g., $\omega = 50$ or 10). Therefore UvsX- ϵ DNA binding appears to be characterized by a moderate level of cooperativity consistent with its filament-forming properties on ssDNA. Similar fits to non-stoichiometric binding isotherms obtained at other salt concentrations (Table 3) also gave ω -values consistently around 100. Note that for both NaCl and KOAc the K parameter decreases markedly with increasing salt concen-

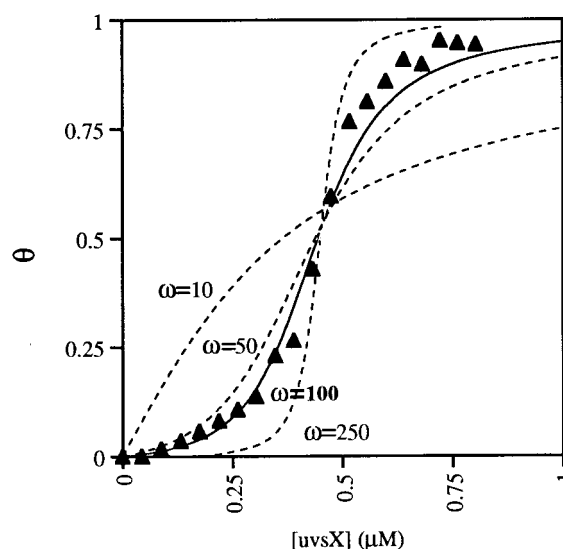


Figure 4. Example of titration data fit to theoretical binding isotherm; method used to estimate K and ω values at a given salt concentration. Data set shown is from titration performed in chloride buffer plus 150 mM NaCl (identical experiment shown in Figure 3). The product $K\omega$ (here equal to $4.3 \times 10^6 M^{-1}$) was calculated from the midpoint of the titration ($\theta = 0.5$) as described in the text. The parameters K and ω were then varied reciprocally while holding the product $K\omega$ constant to generate the theoretical isotherms shown. The value of the binding-site size parameter was set at $n = 4$ for each isotherm. Broken lines represent theoretical isotherms with $K = 4.3 \times 10^5 M^{-1}$, $\omega = 10$; $K = 8.6 \times 10^4 M^{-1}$, $\omega = 50$; and $K = 1.7 \times 10^4 M^{-1}$, $\omega = 250$ respectively, none of which fits the data adequately. The best fit to the data (continuous line) was obtained when parameter values were set at $n = 4$, $K = 4.3 \times 10^4 M^{-1}$, and $\omega = 100$, respectively. Similar visual-fitting procedures were used to generate all of the “best fit” isotherms drawn in Figure 3, and to estimate all of the K and ω parameter values listed in Table 3. Note that for all fits and data sets, greatest weight was given to the data in the lower half of the titration curve, e.g., for $\theta < 0.5$.

tration ([salt]), while the ω parameter is essentially independent of [salt] (Table 3). The salt dependence of K is addressed in the next section. Finally, the K and ω parameters shown in Table 3 were used to generate the theoretical isotherms shown as continuous lines in Figure 3, representing the best visual fits to each of the data sets.

Analysis of salt effects on K

The number of ionic interactions involved in UvsX- ϵ DNA complex formation may be estimated from the slope of a log/log plot of K versus [salt] according to the equation:

$$\frac{d \log(K_{\text{Mg-corr}})}{d \log[\text{salt}]} = -m'\psi - a$$

where $K_{\text{Mg-corr}}$ is the association constant at a given salt concentration corrected for the effects of Mg^{2+} , m' is the number of ionic contacts between protein and DNA phosphate residues, ψ is the fraction of ions thermodynamically bound to the DNA (equal to 0.71 for ssDNA), and a is the number of anions displaced by protein-DNA binding (Record *et al.*, 1976, 1977; deHaseth *et al.*, 1977a,b; Menetski & Kowalczykowski, 1985). The K -values determined in the previous section were corrected for the effects of Mg^{2+} as described by Menetski & Kowalczykowski (1985) based upon equations (2) and (3) of Record *et al.* (1977). The Mg^{2+} -corrected ($K_{\text{Mg-corr}}$) values are listed in Table 3, and are used in the plots shown in Figure 5. This correction is

necessary since Mg^{2+} competes with monovalent cations for binding to nucleic acids and therefore introduces curvature into the $\log K$ versus $\log[M^+]$ plots such as those shown in Figure 5. Note that the Mg^{2+} corrections of the NaCl data are larger proportionally than those of the KOAc data (Table 3), due to the smaller concentrations of the monovalent cations in the former. If we assume a maximum value of $m' = 4$, since UvsX has a binding-site size of $n = 4$, and if we also assume no contribution of anion displacement to the slope (i.e., $a = 0$), then the maximum magnitude that $d \log(K_{\text{Mg-corr}})/d \log[\text{salt}]$ could have is -2.84 . Figure 5 shows that the experimental slopes obtained with the NaCl and KOAc data are -3.12 and -2.92 , respectively. These values do not differ significantly from the theoretical case where $a = 0$, suggesting that anion displacement makes a minimal contribution to the salt effect on K . However, this model is contradicted by the salt-back titration data in Figure 2, which show a strong counterion effect. Therefore a more likely explanation of the slopes calculated in Figure 5 is that UvsX protein makes less than the maximum of four contacts with ssDNA phosphate groups per monomer, and therefore the value of m' must be less than or equal to 3; the remaining magnitude of the slope would thus be accounted for by anion displacement, i.e., the value of a must be greater than or equal to one anion displaced per UvsX monomer. Note that extrapolation of the lines fit to the NaCl and KOAc data in Figure 5 predict that at equivalent salt concentrations, the magnesium-corrected K -value is greater with acetate as the counterion than with chloride, consistent with a strong anion displacement component in UvsX- ϵ DNA binding and with the strengths of ionic interactions predicted by the Hoffmeister series (von Hippel & Schleich, 1969).

Discussion

The fluorescence studies described here support the following conclusions. (1) UvsX protein binds preferentially to an ϵ DNA versus a native ssDNA lattice with a binding-site size of $n = 4$ nucleotides per monomer. (2) The salt-stability of the UvsX- ϵ DNA complex at equilibrium is greatly enhanced by the addition of the non-hydrolyzable ATP analog ATP γ S, but is unaffected by ADP. (3) UvsX binds to ϵ DNA with moderate cooperativity ($\omega = 100$), and this cooperativity is unaltered by salt conditions. (4) The UvsX- ϵ DNA association constant is salt sensitive and includes both cation and anion effects on K . These results allow detailed comparisons between the ssDNA-binding properties of UvsX, RecA protein (its *E. coli* homolog), and T4 recombination factors UvsY and gp32, which we outline below.

It is well established that the UvsX protein binds stoichiometrically and in a non-sequence-specific manner to ssDNA, forming filaments which hydrolyze ATP and catalyze the synapsis and branch-

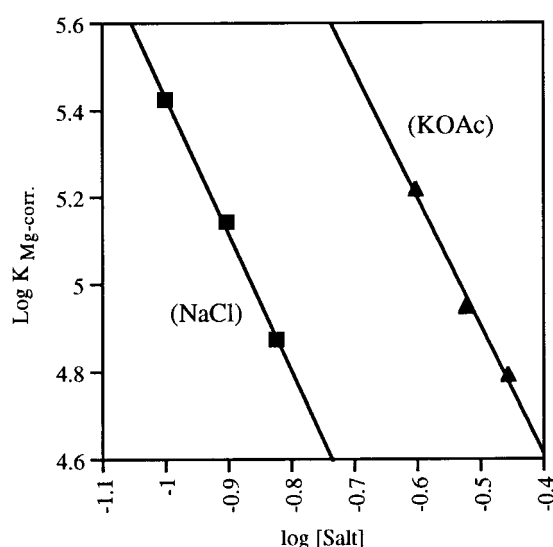


Figure 5. Ion effects on the association constant for UvsX- ϵ DNA interactions. The $K_{\text{Mg-corr}}$ values (from Table 3) represent K values at given NaCl or KOAc concentrations corrected for the effects of Mg^{2+} as described in the text. ■, data from titrations at variable [NaCl]; ▲, data from titrations at variable [KOAc]. The data in each series have been fitted to a least-squares line yielding slopes of -3.12 and -2.93 for the NaCl and KOAc data, respectively. See the text for details and discussion.

migration steps of homologous recombination. By rigorously establishing the binding-site size of UvsX on ϵ DNA to be four nucleotide residues, we have revealed or confirmed several important aspects of the UvsX-ssDNA structure and function: First, that UvsX and the *E. coli* RecA protein have nearly identical stoichiometries for ssDNA binding (Morrical & Cox, 1985), consistent with the similarities in the morphologies of the two presynaptic filaments as revealed by electron microscopy (Griffith & Formosa, 1985; Yu & Egleman, 1993). Second, that UvsX has an identical binding-site size, on a monomer basis, with the T4 UvsY protein, which is the major recombination accessory protein responsible for nucleating the assembly of and stabilizing UvsX-ssDNA presynaptic filaments both *in vitro* and *in vivo* (Sweezy & Morrical, 1997). Previous studies have suggested that UvsX and UvsY may form a 1:1 stoichiometric complex on ssDNA (Kodadek *et al.*, 1989); their identical binding-site sizes on ssDNA appear to support this notion.

In addition to binding-site size, UvsX shares several other ssDNA-binding properties in common with the *E. coli* RecA protein: one of these is its enhanced affinity for the etheno-modified form of random-sequence ssDNA, a property which has been noted in several other proteins with ssDNA-binding activity including RecA, gp32, and UvsY (Menetski & Kowalczykowski, 1985; Kowalczykowski *et al.*, 1981; Sweezy & Morrical, 1997). A second common feature is cooperativity. Previous studies have noted the cooperativity of UvsX-ssDNA interactions by indirect means (Yonesaki & Minagawa, 1985; Formosa & Alberts, 1986), and in fact positive cooperativity is an expected property of this filament-forming protein. Our data establish that the cooperativity parameter (ω) for UvsX protein binding to an ϵ DNA lattice has a value on the order of 100, which is a moderate level of cooperativity for a protein with a small binding-site size ($n = 4$). The ω parameter for the RecA protein, determined by the same method, is approximately 50 (Menetski & Kowalczykowski, 1985). Therefore UvsX and RecA both express similar, moderate levels of cooperativity in their interactions with single-stranded polynucleotides, consistent with their mutual ability to form relatively long nucleoprotein filaments on ssDNA. A third common feature is the ability of ATP γ S to induce a high-affinity ssDNA-binding mode in both proteins. Menetski & Kowalczykowski (1985) argued that the ATP γ S-induced binding mode mimicks a conformation of the RecA protein that forms transiently in the ATP-bound form of RecA-ssDNA prior to the hydrolysis of ATP. It appears likely that a similar model applies to UvsX protein; however, our inability to observe any enhancement in the salt stability of UvsX- ϵ DNA complexes under steady-state conditions for ATP hydrolysis (Figure 2; Table 2) suggests that other factors in addition to ATP are necessary to induce high-affinity binding of UvsX to ssDNA. As mentioned in

Results, this may be a result of the more rapid catalytic turnover of the UvsX ATPase activity compared to RecA (Formosa & Alberts, 1986; Kodadek, 1990). This observation leads us to a discussion of two of the major differences we observe between the UvsX and RecA proteins: (1) the lack of any effect of ADP on UvsX-ssDNA interactions, and (2) the requirement by UvsX protein for an additional protein factor (the T4 UvsY protein) for the stability of presynaptic filaments.

The lack of any observable effect of ADP on UvsX- ϵ DNA complex stability is surprising given the strong destabilizing effect of this nucleotide on RecA- ϵ DNA complexes (Menetski & Kowalczykowski, 1985). Destabilization of RecA-ssDNA has been proposed as a mechanism to remove RecA from ssDNA following ATP hydrolysis (Menetski & Kowalczykowski, 1985; Cox *et al.*, 1983), possibly leading to a dynamic treadmilling of the RecA-ssDNA filament (Lindsley & Cox, 1990). Similarly, ATP hydrolysis has been proposed to set up a treadmilling reaction of UvsX protein on ssDNA by creating separate ATP- and ADP-capped filament ends with different stabilities (Kodadek *et al.*, 1988). However, our observations that (1) ADP fails to destabilize UvsX- ϵ DNA and (2) ATP provides no measurable stability advantage to UvsX- ϵ DNA at steady-state both argue that ATP hydrolysis plays a less direct role in UvsX-ssDNA filament dynamics than is the case for the RecA protein. The most likely explanation for this difference between UvsX and RecA is the involvement of the T4 UvsY protein in presynapsis. UvsY is clearly a major controlling factor in the stability of UvsX-ssDNA complexes; protein-protein interactions between UvsX and UvsY render UvsX-ssDNA filaments highly resistant to both salt and dilution of the complex (Kodadek *et al.*, 1989; Kodadek, 1990; Morrical & Alberts, 1990; Yonesaki & Minagawa, 1989), presumably by blocking dissociation of UvsX from the ssDNA. Thus UvsY may obviate the need for a type of UvsX-release mechanism based on the nucleotide hydrolytic cycle; however, this leaves open the question of how, or even whether, mature UvsX-UvsY-ssDNA presynaptic filaments are capable of dissociation and reassembly under physiological conditions.

The results presented here complement detailed thermodynamic analyses of UvsY-ssDNA interactions (Sweezy & Morrical, 1997) and gp32-ssDNA interactions (Kowalczykowski *et al.*, 1981; Newport *et al.*, 1981) resulting in a detailed understanding of how these three T4 recombination proteins bind independently to the ssDNA lattice under defined conditions *in vitro*. Comparative binding parameters estimated for all three proteins are listed in Table 4. By extrapolating the $\log K$ versus $\log[\text{salt}]$ curves to 200 mM NaCl, comparisons can be made to the ssDNA interactions of UvsY and gp32 under these salt conditions which approximate the physiological ionic strength. For native ssDNA, the estimated (Mg^{2+} corrected) $K\omega$ value for gp32 is $1 \times 10^9 \text{ M}^{-1}$ and that for UvsY is

Table 4. Comparison of ssDNA-binding parameters (Mg^{2+} corrected) estimated for T4 recombination proteins UvsX, UvsY, and gp32 at 200 mM NaCl

Protein	<i>n</i>	$K_{Mg-corr}\omega$ (M^{-1})	$K_{Mg-corr}$ (M^{-1})	ω	References
UvsX	4	$<3.1 \times 10^6$	$<3.1 \times 10^4$	100	This study
UvsY	4	$\sim 1 \times 10^7$	$\sim 1 \times 10^7$	1	Sweezy & Morrical (1997)
gp32	7-10	$\sim 1 \times 10^9$	$<1 \times 10^6$	>1000	Kowalczykowski <i>et al.</i> (1981) Newport <i>et al.</i> (1981) Kowalczykowski <i>et al.</i> (1986)

See the text for details.

$1 \times 10^7 M^{-1}$ (Newport *et al.*, 1981; Sweezy & Morrical, 1997). The 200 mM NaCl estimate for UvsX is $K\omega < 3.1 \times 10^6 M^{-1}$ ($K\omega$ for native ssDNA must be less than this value derived for UvsX- ϵ ssDNA, since we find that UvsX binds much more strongly to ϵ ssDNA than to native ssDNA). Thus the UvsX-ssDNA interaction is clearly much weaker than the other two interactions. Combined with the apparent lack of a significant stabilizing effect of ATP on UvsX-ssDNA interactions, the relatively weak observed association of UvsX with ssDNA argues that UvsX-UvsY protein-protein interactions are necessary to stabilize UvsX on the ssDNA as well as to displace tightly bound gp32. One possible mechanism of UvsY action would be for UvsY to trap and prolong the lifetime of an ATP-bound, high-affinity form of the UvsX protein on the ssDNA, e.g., a conformation of UvsX related to that induced by ATP γ S. While other models are surely possible, this one should be testable by both steady-state and pre-steady-state kinetics experiments of the UvsX ATPase activity.

Other binding parameter comparisons between UvsX, UvsY, and gp32 (summarized in Table 4) are intriguing, but their functional significance is still open to interpretation: binding site size estimates for the three proteins include $n = 7$ to 10 for gp32 and $n = 4$ for both UvsX and UvsY. As mentioned previously, these relationships have been used to argue for a 1:1 stoichiometry of ssDNA-bound UvsX and UvsY in the mature presynaptic filament (Kodadek *et al.*, 1989; Sweezy & Morrical, 1997), as well as for a 1:2 stoichiometric relationship between gp32 and UvsY monomers within an intermediate gp32-UvsY-ssDNA complex (Sweezy & Morrical, 1998). Note that the cooperativity parameter of UvsX ($\omega = 100$) is intermediate between the high cooperativity case of gp32 ($\omega > 1000$) and the no cooperativity case of UvsY ($\omega = 1$) which instead appears to bind to ssDNA in isolated hexameric units (Beernink & Morrical, 1998). It is intriguing that the activation of the highly filamentous T4 recombination machinery requires proteins which exhibit the entire range of possible cooperativities, from the complete absence of cooperativity with UvsY, to the essentially "unlimited" type of cooperativity seen with gp32.

Observations in the T4 system are finding parallels in the recombination systems of both prokaryotic and eukaryotic cells, exemplified by recombination studies with RecA of *E. coli* and

Rad51 of *Saccharomyces cerevisiae*. The necessity to recombine, replicate and repair DNA is as universal as the lattice itself, and multiprotein reactions among helix-destabilizing proteins, recombinases and recombination accessory proteins also appear to be conserved. The T4 model of increased stability and loading of UvsX by UvsY onto gp32-coated ssDNA is analogously illustrated in *E. coli* by the RecO/R stimulation of RecA toward SSB-coated ssDNA (Umezū *et al.*, 1993; Umezū & Kolodner 1994; Shan *et al.*, 1997) and in *S. cerevisiae* by the Rad52, Rad55/57 stimulation of Rad51 toward RP-A-coated ssDNA (Firmenich *et al.*, 1995; Sung, 1997a,b; Benson *et al.*, 1998; New *et al.*, 1998; Shinohara & Ogawa, 1998). The new information we have gleaned here regarding the UvsX-ssDNA interaction complements our understanding of the other T4 presynapsis proteins, UvsY and gp32, and provides a framework for experiments underway to decipher interactions responsible for mature presynaptic filament formation and subsequent recombinational events. This detailed biochemical characterization may provide models for some of the equivalent interactions of homologous recombination proteins in other systems.

Materials and Methods

Reagents and buffers

All the chemicals used here were reagent grade, and all aqueous buffers and solutions were made with filtered, deionized and glass-distilled water. All buffers and reagents used for the fluorescence studies were sterile filtered at 0.45 microns. ADP, ATP γ S, phosphoenolpyruvate (PEP) and pyruvate kinase (PK) were purchased from Sigma. ATP was purchased from Pharmacia. All other chemicals, biochemicals, enzymes, and resins were purchased from Sigma unless otherwise noted. Chloride-based fluorescence buffers included 20 mM Tris-HCl (pH 7.4), 3 mM magnesium chloride, and variable NaCl concentrations. Acetate-based fluorescence buffers included 20 mM Tris-acetate (pH 7.4), 3 mM magnesium acetate, and variable potassium acetate concentrations.

UvsX protein and nucleic acids

The T4 UvsX protein (40 kDa) was purified using a modification of the procedure of Yonesaki & Minagawa (1985), involving successive DEAE-cellulose, hydroxyapatite, and phosphocellulose chromatography steps. UvsX protein purified by this method was determined to

be greater than 98% homogeneous as estimated from samples run on SDS-polyacrylamide gels stained with Coomassie brilliant blue. Final, purified fractions from the phosphocellulose column were dialyzed into UvsX storage buffer (UvsX SB) containing 20 mM Tris-HCl (pH 7.4), 1 mM 2-mercaptoethanol, 0.2 mM EDTA, 100 mM NaCl, and 10% (w/v) glycerol. Samples in storage buffer were snap frozen in liquid nitrogen and stored at -70°C . The concentration of UvsX protein in stock solutions was determined by the absorbance at 280 nm using an extinction coefficient of $69,760\text{ M}^{-1}\text{ cm}^{-1}$ which was determined from the amino acid sequence of UvsX protein by the method of Gill & von Hippel (1989). Purified UvsX protein was tested for contaminating nuclease activities as described (Morrical *et al.*, 1994), and found to be nuclease-free by these criteria.

Circular single-stranded DNA from bacteriophage M13mp19 was isolated from purified phage particles as described by Yamamoto *et al.* (1970) and Miller (1987). This ssDNA was used to make the fluorescent ϵ DNA derivative according to the method of Menetski & Kowalczykowski (1985). The integrity of the ϵ DNA was verified by agarose gel electrophoresis as described (Sweezy & Morrical, 1997); the ϵ DNA had a slightly slower mobility than unmodified ssDNA, and appeared unfragmented. The concentrations of both etheno-modified and unmodified ssDNA species were determined by the phosphate-ash method (Ames, 1966; Lefebvre & Morrical, 1997). All ϵ DNA and ssDNA concentrations are expressed in terms of nucleotide residues.

Fluorescence assays

All fluorescence data were obtained on an SLM model 8000 fluorimeter and collected with Olis software taking ratiometric readings between the sample cell and a rhodamine-containing standard. All fluorescence measurements were made by monitoring a 405 nm emission wavelength through an 8 nm bandpass following excitation at 300 nm. Photomultiplier voltage was kept constant during all titrations, and settings were selected so that maximal fluorescence readings were less than 80% of the detector's linear range. All data points consisted of an average of at least five fluorimeter readings, set on continuous mode. All titrations were performed at a constant temperature of 25°C , maintained by a circulating water bath connected to the cuvette block. All reactions contained a 2.0 ml starting volume with the appropriate chloride or acetate buffer and other components (described below) prior to the addition of titrant. Starting samples were preincubated for five minutes to allow thermal equilibration, then 5 μl increments of titrant (either salt or protein solution) were added and allowed to mix and equilibrate by magnetic stirring for one minute prior to taking each fluorescence reading.

When necessary, fluorescence data were corrected for sample dilution, intrinsic protein fluorescence, fluorescence of free ϵ DNA, and/or inner filter effects as described (Sweezy & Morrical, 1997; Lefebvre & Morrical, 1997; Birdsall *et al.*, 1983). The intrinsic fluorescence of UvsX protein was determined in protein-only control titrations, and was found to vary linearly with UvsX concentration in all cases. No correction was made for photobleaching since negligible decreases in ϵ DNA or intrinsic protein fluorescence were observed during prolonged exposure to the excitation beam. However, slits remained closed during sample addition and equili-

bration to minimize possible photobleaching of the sample.

Three types of fluorescence titration experiments were performed: (1) forward titrations of ϵ DNA lattice with UvsX protein under either stoichiometric (tight-binding) or non-stoichiometric (weak-binding) conditions, monitoring UvsX- ϵ DNA complex formation by increasing ϵ DNA fluorescence; (2) competition titrations i.e., forward titrations of UvsX protein into a mixture of ϵ DNA and unmodified ssDNA, again monitoring UvsX- ϵ DNA complex formation by increasing ϵ DNA fluorescence; and (3) salt-back titrations of pre-formed UvsX- ϵ DNA complexes with NaCl or KOAc, monitoring complex dissociation by decreasing ϵ DNA fluorescence. These methods are described by Kowalczykowski *et al.* (1986), Sweezy & Morrical (1997), and Lefebvre & Morrical (1997). Specific reactions conditions are described below.

Forward stoichiometric titrations to determine the binding-site size (n) parameter of UvsX protein on ϵ DNA were performed in both chloride and acetate buffer systems containing 50 mM NaCl or KOAc, respectively. Fixed amounts of ϵ DNA (1.05, 2.1, and 3.2 μM) were equilibrated in chloride or acetate buffer, then 5 μl additions of UvsX from a concentrated stock (UvsX storage buffer with ionic strength adjusted to match that of the starting buffer) were added and the fluorescence was monitored as described above. Controls included titrations lacking ϵ DNA to monitor intrinsic protein fluorescence, and substitution of UvsX stock buffer in lieu of UvsX to monitor the effects of dilution. Some reactions contained ADP (200 or 500 μM), ATP γ S (50 or 100 μM) or an ATP-regeneration system consisting of 100 μM ATP, 2 mM PEP, and 2.6 units/ml pyruvate kinase. The appropriate mock titrations were performed for each. Competition titrations were performed identically with the forward stoichiometric titrations in the absence of nucleotide cofactors, with the exception that starting mixtures contained equimolar amounts of ϵ DNA and unmodified M13mp19 ssDNA.

Forward non-stoichiometric titrations to determine the affinity (K) and cooperativity (ω) parameters of UvsX- ϵ DNA interactions were performed similarly to the stoichiometric titrations described above, except that NaCl or KOAc concentrations were adjusted to provide weak-binding conditions ($[\text{NaCl}] = 100$ to 150 mM , or $[\text{KOAc}] = 250$ to 350 mM , determined independently in salt-back titrations). All reactions had an ϵ DNA concentration of 2.1 μM . Fluorescence data corrected for dilution, intrinsic protein fluorescence, and inner filter effects were converted to terms of fractional saturation of the lattice (θ) by subtracting the fluorescence of free ϵ DNA (determined independently at a given salt concentration) then normalizing the plateau fluorescence signal to a value of $\theta = 1$, as described by Lefebvre & Morrical (1997). Binding curves consisting of experimental values of θ were fit visually to theoretical isotherms generated from the equations of McGhee & von Hippel (1974, 1976) as described (Kowalczykowski *et al.*, 1986) by (1) calculating the value of the product $K\omega$ from the midpoint of the titration (i.e., $\theta = 0.5$) according to the relationship $1/K\omega = L_{T,0.5} - N/2n$, where $L_{T,0.5}$ equals the total protein concentration at the titration midpoint, N equals the DNA concentration in nucleotides, and n equals the binding-site size, and (2) varying the values of K and ω reciprocally while holding their product constant until a reasonable visual fit was obtained.

Salt-back titrations employed a 1:1 stoichiometric complex of UvsX protein and ϵ DNA (0.53 μM and 2.1 μM ,

respectively, assuming a binding-site size of $n = 4$ nucleotide residues for the UvsX protein; see Results), pre-equilibrated either in chloride or acetate buffer containing a starting concentration of 50 mM NaCl or 200 mM KOAc, respectively. Salt, from either a 5 M NaCl or a 5 M KOAc stock was added to the cuvette by 5 μ l additions, and fluorescence was monitored as described above. Some reaction mixtures contained ADP, ATP γ S, or ATP plus a regenerating system, each at the concentrations stated above for the forward titrations. Each salt-back titration was accompanied by control titrations in which either ϵ DNA or UvsX protein was omitted, to independently assess the effects of changing salt concentration on ϵ DNA and intrinsic protein fluorescence.

Acknowledgments

The authors thank Mark Sweezy and Steven Lefebvre of this laboratory for help with preliminary fluorescence work. We also thank David Yassa and Yujie Ma of this laboratory for technical assistance. This work was supported by research grant #GM48847 from the National Institutes of Health.

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Edited by M. Gottesman

(Received 30 March 1998; received in revised form 30 July 1998; accepted 31 July 1998)